# HelSpin — User Manual

Version 0.9.20

Compare Bruker NMR spectra and build publication figures.

Contact: iwai@ligsciss.com

# 1. What HelSpin is for

You have a folder of Bruker datasets and a question that needs several of them side by side: did the reaction go, does this batch match the reference, which of six conditions gave the cleanest product. HelSpin opens them together, lets you line them up, and exports the result at publication quality.

It reads Bruker data directly from wherever it already lives, including a mounted spectrometer share. **It never writes to your data.** Everything you adjust — scale, position, colour, zoom — is display state, held in HelSpin and saved in its own session files.

# 2. Installing

On macOS, if you have `HelSpin-<version>-<arch>.dmg`: open it and drag HelSpin to Applications. No Python or conda is needed. Check the architecture in the filename: `arm64` for Apple Silicon, `x86_64` for Intel; they are not interchangeable.

If the image is not notarised by Apple, the first launch is blocked ("Apple could not verify “HelSpin” is free of malware"). Click **Done**, then **System Settings** → **Privacy & Security** → **Open Anyway** (once only). On macOS 15 and later, right-click → Open no longer gets past this. `INSTALL.md` has the details and a Terminal alternative.

Otherwise, and on Windows and Linux, see `INSTALL.md`. Linux needs one system library (`libxcb-cursor0`) before the first run.

# 3. Getting started

## Adding a data root

**File** → **Add Data Root…**, then pick the folder your datasets sit under. Point it at a directory that *contains* sample folders, not at a single dataset. HelSpin walks down looking for the Bruker structure (a sample folder holding numbered experiment folders -- numbered folders with `acqus`, `fid`, `ser` or `pdata` in them) and shows what it finds. Other numbered folders, such as a year (`2026`) or an instrument (`600`), are searched through, not taken for samples.

You can add more than one root — a network share and a local archive, say — and they persist between sessions.

Symlinks are followed, so a root made of links to the real instrument mounts works normally. Links pointing back at a parent are detected and not followed twice, so a loop cannot hang the scan.

## The browser

The left panel is the tree: root → sample → experiment. Samples are listed directly under their data root, whatever folders they sit in; hover over a sample to see its folder (e.g. `NMR600_2026/data/ETHZ/Shared/NRAS`).

Type in the filter box to narrow by sample name, pulse programme, or any folder name between the data root and the sample — `ETHZ` shows every sample below an `ETHZ` folder, `Shared/NRAS` those below that one. The filter reaches samples that have not been expanded yet.

The tree does **not** watch the filesystem. Network shares make that unreliable, so refreshing is explicit: right-click a node to refresh it, or **Refresh All** / **F5**. Refreshing a data root finds new and removed samples, and also looks inside known samples — expanded or not — for new experiments and for experiments processed since they were last seen. That second part runs in the background ("Indexing in background…").

At start-up HelSpin also re-checks, by itself, samples changed in the last two weeks that still have experiments in progress, so freshly processed data usually appears without a refresh.

A data root holding more than 100,000 samples is cut off at that point: its row says so, and HelSpin warns once in a dialog -- also when a root comes within 10% of the limit. Add a more specific folder as its own data root instead.

## Opening spectra

Drag a dataset onto the canvas, or double-click it. Repeat to add more. Each one becomes a *trace* listed in the right-hand panel.

If a dataset does not appear, ask HelSpin what it sees on disk:

```
helspin --check /path/to/SAMPLE
```

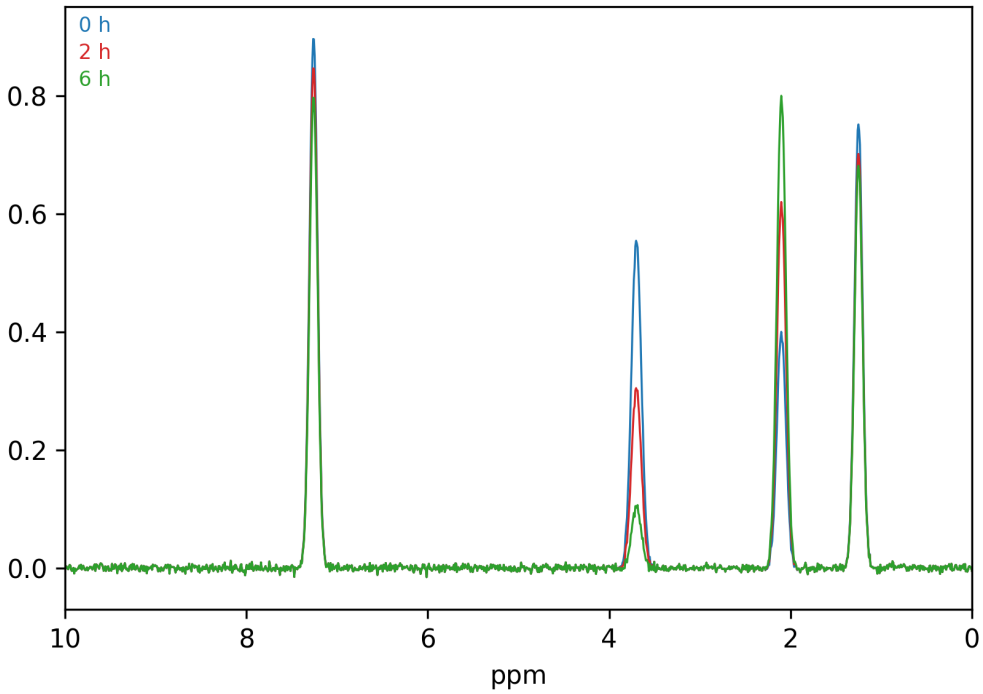
That reports, experiment by experiment, whether a plottable spectrum is actually there.

### The toolbar

**Quit** is the red button at the right end of the toolbar. It always stays there, whatever the window width. When the window is too narrow for every button, the other buttons get smaller text first; only if that is still not enough do the rightmost of them move behind **»** (click it to reach them). Widen the window and they return to normal size.

## 4. Arrangements: overlay and stacked

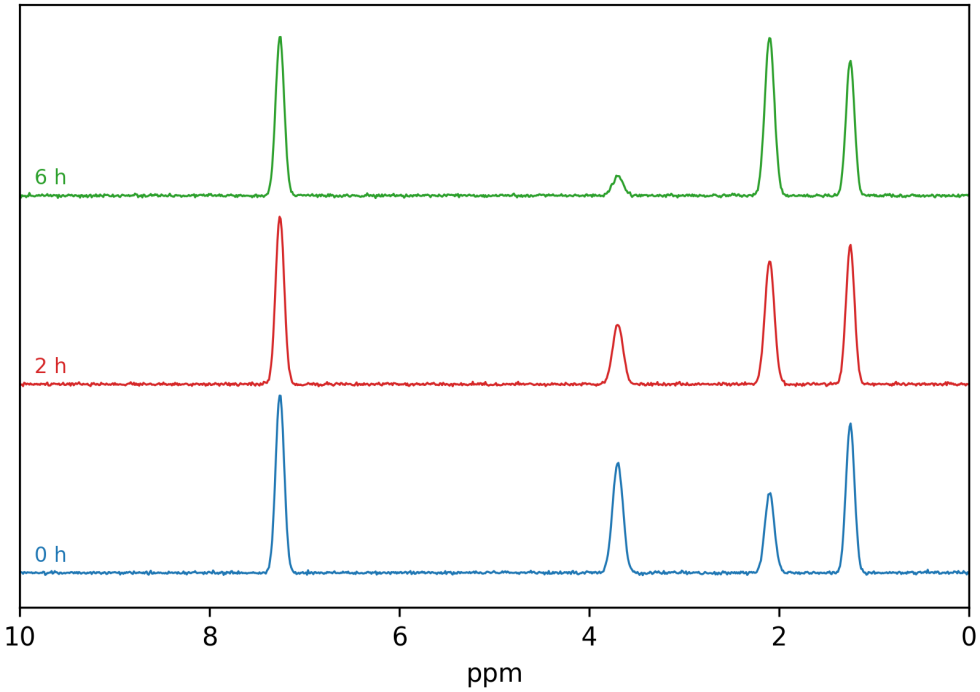
**Overlay** draws every spectrum on one set of axes, sharing a baseline. Use it to compare peak positions and relative intensities directly.




Overlay

**Stacked** gives each spectrum its own lane, evenly spaced. Use it when the traces would otherwise obscure each other, and for showing a progression.

In a stack each spectrum's name sits **beside its own trace**, so a reader does not have to match colours to work out which is which. The name is anchored to the lane's baseline, so it stays put when you scale or magnify the spectrum, and follows the lane when you zoom. Drag a name to reposition it; **Reset label positions** puts them back.




Stacked

Switch with the arrangement control. The frame is re-fitted on switching, because offsets that suit one layout rarely suit the other.

# 5. Adjusting a spectrum

Select a trace in the right-hand panel first — adjustments apply to the selection.

## Y scale — the wheel

With both zoom toggles off, the **mouse wheel over the plot scales the selected spectrum** vertically. This changes that spectrum's height relative to the others, which is what you want when one sample is far more concentrated than another.

Nothing happens if no trace is selected. That is deliberate: guessing which spectrum you meant would be worse than asking you to click one.

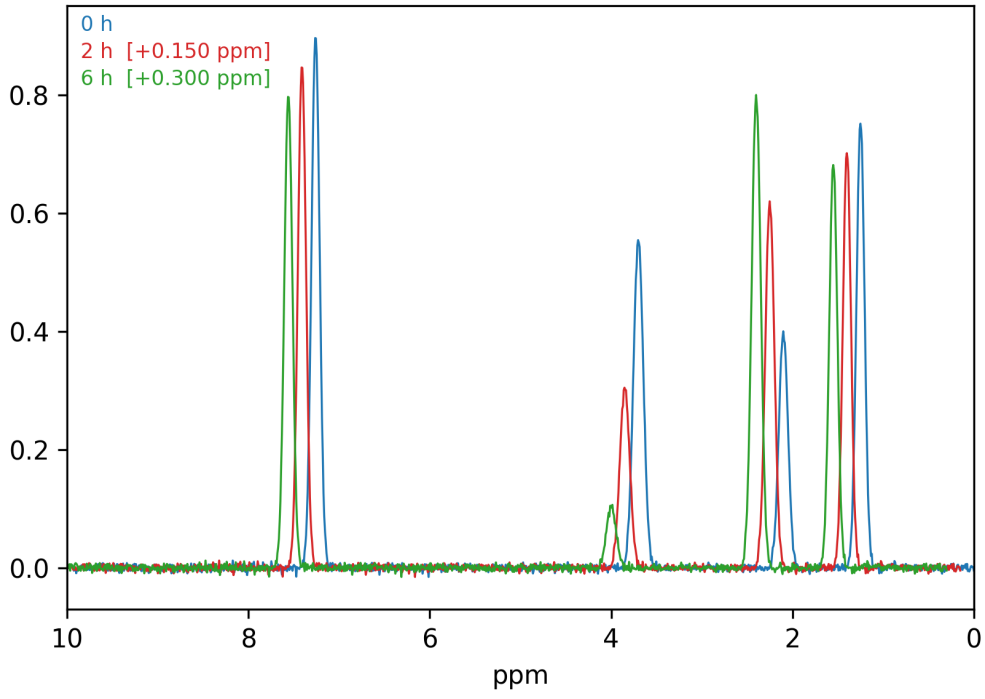
## Y offset

Moves the selected spectrum up or down. Pure translation — it does not rescale anything and does not re-fit the view. You can also drag the selected spectrum on the plot; a plain click (without moving a few pixels) never moves it, and a drag is one step for Edit → Undo.

**To bottom** drops the selected spectrum to the baseline; **Bottom all** does every spectrum at once.

## X offset — shifting along ppm

**Spectrum list** → **X offset**, in ppm. Shifts the selected spectrum horizontally. Two uses: correcting spectra that were referenced slightly differently, and skewing a stack into a cascade.




X offset

The step size and range come from the spectrum's own width, so the control behaves sensibly for a $^{1}$H spectrum spanning 12 ppm and a $^{13}$C one spanning 200. You can shift up to half a spectrum width in either direction.

> **A shifted spectrum says so.** An X offset moves a trace along the chemical shift axis, so its peaks no longer read at their true values. HelSpin marks any shifted trace in the spectrum list *and on the plot label* — `2 h [+0.150 ppm]` — so a figure never quietly claims a peak sits somewhere it does not. The marker disappears when the shift returns to zero. The underlying data is never modified. Peak readout always reports true, unshifted ppm, and **Clear X offsets** returns everything to its real chemical shift in one action.

## Appearance

Colour, line width and line style are per-spectrum, in the spectrum list. Defaults come from the slot palette in Preferences, so the first spectrum you open always looks the same way.

# 6. Zooming

Two independent toggles in the bottom bar: **X zoom** and **Y zoom**. They are not exclusive — with both on, the wheel zooms both axes at once.

| Toggle | What the wheel does |
|---|---|
| Both off | Scales the **selected spectrum** vertically |
| X zoom | Zooms the ppm axis about the cursor |
| Y zoom (overlay) | Zooms the intensity axis about the cursor |
| Y zoom (stacked) | **Magnifies all spectra**, leaving the lanes where they are |
| Both on | Zooms both axes together |

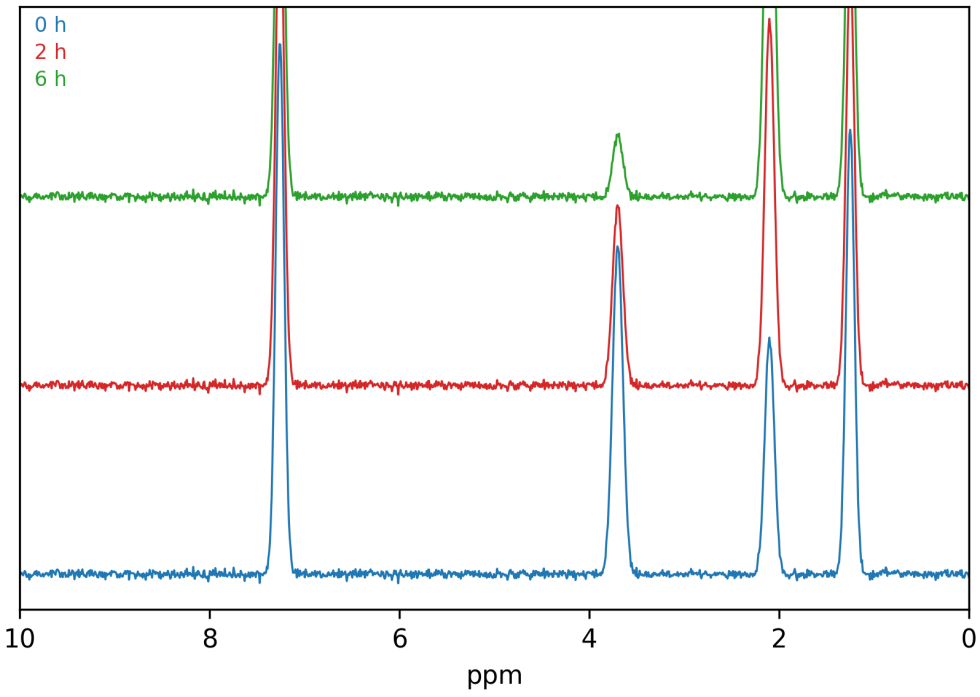
Zooming keeps the point under the cursor fixed, so a peak of interest does not walk off the edge.

With **X zoom** on, you can also drag a box on the plot to zoom into it. A click is never a zoom: the pointer has to move a few pixels first, so clicking the plot keeps the range you have. Dragging past the frame zooms up to the edge. A box zoom is one step for Edit → Undo. With **Y zoom** also on, the box sets the vertical range too (overlay only — in a stack the box sets the ppm range and leaves the lanes alone).

## Y zoom in stacked mode is different, on purpose

In a stack, the layout *is* a set of lanes at fixed baselines. Narrowing the vertical window there would not magnify the stack — it would crop it, pushing whole spectra off the canvas.

So in stacked mode the wheel **magnifies the traces** and leaves the baselines exactly where they are. Peaks grow in place, overflow their lane, and can run off the top of the canvas.




Stacked and magnified

That overflow is intended. Bringing up a weak signal beside a strong one can need magnification of many orders of magnitude, so **there is no limit** in either direction.

## Getting back

- **Fit Y** — re-frames the vertical axis and clears any Y zoom or magnification.
- **Full** — shows everything, in every axis.
- **Ctrl+Z** — undoes it, including Fit Y.

A zoom is sticky: it survives loading another spectrum, removing one, changing the ppm window and redrawing. Turning the toggle off does not undo it. Only Fit Y, Full, reset or undo clear it.

# 7. Setting the ppm window

The bottom bar takes a typed range. ppm axes descend, so the **left** box is the higher value. **Enter** or **Apply** sets it; **Full** clears it.

Moving between the two boxes applies nothing, so you can type both before applying. If you click the plot (or anything else in the window) without applying, the boxes go back to the range on the plot — the plot keeps the range you last applied.

Recently used ranges are remembered **between runs**, so returning to a window you use often is one click. The list starts seeded with three common windows rather than empty:

| Range | Typical use |
|---|---|
| 0 to 12 ppm | the standard $^{1}$H sweep |
| -1 to 13 ppm | the same, with a margin at each end |
| 5 to 13 ppm | downfield only: aromatics, amides |

Ranges you apply are added to the front, so the list becomes yours with use.

Narrowing the window re-fits the vertical frame to the data actually in view — otherwise zooming into a quiet region would keep the scale set by peaks outside it and show you a flat line. An explicit Y zoom outranks that and is kept.

# 8. Undo

**Ctrl+Z** / **Ctrl+Y** (or **Ctrl+Shift+Z**). Forty steps.

Undo covers display state — scales, offsets, zooms, colours, adding and removing spectra. It never re-reads a file, so undoing a change costs nothing even with data on a slow share, and undoing a removal brings the spectrum back instantly with its data intact.

Typing into a spin box is one undo step, not one per keystroke.

# 9. Sessions

**File** → **Save Session…** writes everything about the current view: which datasets are open, and every scale, offset, colour, zoom and window. **File** → **Open Session…** restores it.

Sessions store *paths*, not spectra. If the underlying data has moved, the session cannot find it — keep the data root stable, or re-add the root.

Restoring clears the undo history, so an opened session cannot be undone into the state that preceded it.

# 10. Exporting a figure

Right-click the canvas → **Copy image** puts the figure on the clipboard, to paste straight into PowerPoint, Keynote, Word, a mail or a chat. Also **Edit** → **Copy Image** (Ctrl+Shift+C; on a Mac Cmd+Shift+C). It is the same picture as a saved PNG: 300 dpi, white background, no cursor crosshair.

Right-click the canvas → **Save image…**. Formats: PNG, JPEG, TIFF, SVG, PDF, EPS, PostScript and PowerPoint (one slide). The format follows the extension, or the chosen filter if you type none. SVG, PDF and EPS stay vector with editable text and are what you want for a manuscript.

- **300 dpi** for raster formats, on a white background.
- The exported figure matches what is on screen — same size, same limits, including X offset markers and pinned markers. The cursor crosshair is left out.
- The CEST plots save through exactly the same code and formats (see CEST mode).

# 11. Where HelSpin keeps its files

| What | macOS / Linux | Windows |
|---|---|---|
| Settings | `~/.config/HelSpin/` | registry |
| Index cache, licence | `~/.cache/helspin/` | `%LOCALAPPDATA%\HelSpin\cache` |

On Linux, `XDG_CACHE_HOME` is honoured — worth setting if `/home` is a slow network mount, since the cache belongs on local disk. `HELSPIN_CACHE_DIR` overrides it explicitly on any platform.

Deleting the cache is safe; it is rebuilt on the next scan.

# 12. Troubleshooting

**A sample is missing from the tree.** Ask HelSpin why, from a terminal:

```
helspin --why /path/to/SAMPLE
```

It retraces the explorer's search from your data root to that folder, says at each level what it decided (searched, a sample, too deep, a link...), and whether the sample is in the explorer's saved index -- if not, press Refresh (F5). Any folder below the sample works too, e.g. `.../5/pdata/1`. `helspin --check /path/to/SAMPLE` then reports, experiment by experiment, whether a plottable spectrum is there.

To see what a search of a whole data root finds -- and which folders it could not read -- run the explorer's own search from a terminal:

```
helspin --scan /path/to/DATA_ROOT --find /path/to/SAMPLE
```

It reports how many samples it found in each top-level folder, every folder or entry it could not read (and skipped), and whether the given sample was found. On a large share this takes as long as the explorer's first search. Note that Bruker writes `acqus`, `pdata` and `1r` in lower case and Linux filesystems are case-sensitive, so data copied from Windows by a tool that changed case will not be recognised.

**Permission denied on a share.** Confirm you can read it outside HelSpin first (`ls /mnt/nmr`). An unreadable subtree is stepped over silently by design, so one locked folder does not cost you the rest of the share.

**Linux: the application will not start**, with a Qt platform plugin error. Install `libxcb-cursor0` (Debian/Ubuntu), `xcb-util-cursor` (Fedora/Arch). See INSTALL.md.

**Linux: the window never appears, no error.** Check for a leftover `QT_QPA_PLATFORM=offscreen` in your shell.

**A spectrum vanished after zooming.** Press **Fit Y**.

**The plot looks wrong and you cannot say why. Full**, then **Fit Y**, returns to a neutral view without closing anything.

# 13. Known limitations

Stated plainly, because finding these out mid-figure is worse.

- **Differences and sums ignore an X offset.** If you align two spectra horizontally and then subtract, the subtraction uses their true ppm axes, not the aligned ones.
- **No automatic file-watching.** Refresh is explicit, by design — see §2.
- **The TopSpin bridge is not wired up.** TopSpin identifiers parse, but there is no live connection to a running TopSpin.
- **CEST: no mouse picking of the measured peak.** Type the shift or pick from the Resonances list.
- **CEST: overlays share one "offset from carrier" axis.** Experiments recorded with different O1P do not line up on it.
- **CEST: two-word frequency-list headers** (`sfo hz`, `bf ppm`) are refused.
- **Windows and macOS are not yet verified end to end.** Development and the test suite currently run on Linux. The Windows installer path in particular has not been exercised from start to finish.

# 14. Licence

Free for academic research, teaching and personal use. Commercial use requires a licence — enquiries to **iwai@ligsciss.com**. Full terms in `LICENSE`, and readable in the application at **Help** → **Licence**.

Qt (via PySide6) is LGPLv3 and is packaged one-directory so its shared libraries stay replaceable. nmrglue, NumPy and SciPy are BSD; matplotlib is BSD-style.

# Markers

Right-click the spectrum canvas to pin a vertical or horizontal marker where you clicked, labelled with its value. **Remove nearest marker** drops the closest one; **Clear all markers** empties them.

Markers are held in data coordinates, so they stay on the feature when you pan, zoom or rescale. Their colour, width, dash and opacity are in Preferences, and default to matching the cursor crosshair.

# CEST mode

Toolbar **CEST…**, or File → CEST…. Opens its own window. If exactly one experiment is selected in the browser it is loaded straight away.

The panel handles both Waudby-lab pseudo-2D experiments and picks the tab from PULPROG: `19f_calib_nut` opens the calibration tab, `19f_cest` the Z-spectrum tab.

**Show fitted curves** (in the header) hides the fitted lines on both tabs and leaves the measured points, residuals and reported numbers. Useful for judging a marginal dip against the data rather than against the model drawn through it.

## Using only some rows (from row N to row M)

**Rows:** in the header selects which rows (t1 increments, counted from 1) both tabs use: **from row** N **to row** M, or *to end*. Use it to drop blank or bad rows at the start (*from row 9 to end*) or the end (*from row 1 to row 56*). The default, from row 1 to end, uses every row.

- In CEST the matching saturation offsets are dropped with the rows, so keep at least one $I_0$ reference offset inside the range.
- In the calibration tab the pulse lengths keep their true values: row 9 is still the ninth pulse length, not the first.
- Whatever was already fitted or built is redone at once. The file path line shows e.g. "rows 9–64 of 64". A range that selects nothing falls back to every row, with a note.
- Experiments added for comparison use the same range and their legend entry says so ("[rows 9–64]"). Curves already overlaid keep the rows they were measured with.

## Which data it reads

**Raw `ser` by default.** `xf2` stores SI(F1) rows, and TopSpin defaults SI to a power of two — so a 42-offset series processed at SI(F1) = 32 silently loses ten offsets. The resulting `2rr` is a valid file with no record of the loss. The panel treats the frequency list as the authority on how many rows should exist, and says so in the notes box when pdata disagrees.

Set **Source** to *Processed 2rr* if you want the phasing you set in TopSpin; the panel will still warn if rows are missing. SI(F1) larger than TD(F1) is fine — the surplus rows are blank padding and are trimmed.

To keep TopSpin's own view complete, set `1 SI` to the next power of two above your offset count (64 for 42 offsets) before `xf2`, and accept the blank trailing rows.

## Calibration tab

Choose a peak centre and half-width, then **Fit nutation**. The window is symmetric — centre ± half-width, so 0.03 means a 0.06 ppm span — and the **Window** readout shows the resulting range and its point count. It is applied identically to every row and the result stays signed, because the sign inversion across rows is the measurement.

- **Measurement** — integration is the default and the right choice here. A narrow window works best: on 19F test data, residuals fell from 7.0% at ±0.20 ppm to 3.9% at ±0.03 ppm.
- **Model** — *Automatic* compares a plain sine against a damped one on AICc and reports which won and by how much.
- **Target field** — the field you want. It need not match this calibration's CNST8.

Results end with a **TO SET ON THE SPECTROMETER** table giving the `CNST` value to type for each true field, with the pulse length and power the sequence will derive from it. Lowering `CNST` is the whole adjustment, and in `19f_cest` it is the only one available — that sequence recomputes `plw25` unconditionally, so anything typed into `PLW25` is overwritten when the pulse program compiles. Record the true field in the title, since acqus will show the lowered `CNST` rather than the field you applied.

Results also give the fitted B1 with an error bar, T2 effective, residuals, and two power figures kept deliberately apart: the **correction factor** $(\texttt{nominal/fitted})^2$, which multiplies any nominal power on the same probe and tuning and so transfers to a CEST experiment at a different field; and the absolute power for your target field, which also changes the field and therefore does not transfer. **Export record…** writes a record sheet matching Part 8 of the 19F CEST manual.

## Z-spectrum tab

Saturation offsets are read from the experiment's own `lists/f1/` copy of the FQ1LIST. A bare column of Hz offsets (the usual form) and lists headed by a single keyword — `O1`/`Hz`, `P`/`ppm`, `SFO` (absolute MHz) or `BF` — are understood, as are comments and Windows line endings. Lists with a two-word TopSpin header such as `sfo hz` or `bf ppm` are currently refused with "frequency list unreadable" rather than misread. Without a list the panel says so rather than plotting against row number. A series with fewer rows than offsets is plotted against the offsets it has, with the missing range named; more rows than offsets is refused, since the extras cannot be matched to anything.

- **Measurement** — peak height is the default and usually correct. On 19F test data, repeated reference offsets scattered by 0.007 using height against 0.090 using the integral: a saturated peak leaves a drifting baseline that an integral accumulates without bound, far enough to drive $I/I_0$ negative.
- **Reference from** — offsets at least this far out are averaged to give $I_0$. *Automatic* picks the remote ones. Two or more give a scatter figure, which is a direct measure of the noise on every other point — include a couple of remote offsets in your list for this reason.
- **Fit deepest dip** — a Lorentzian on the points near the dip, reporting its centre in Hz and ppm from the carrier. A dip sitting on the carrier is flagged as likely direct saturation rather than exchange. Every other minimum clearing the sigma threshold is listed with its significance.
- **Error bars from spectrum noise** — one sigma per point, anchored to the spread of the repeated I0 references. A shaded band marks ±1 sigma.
- **Subtract fitted dip** — removes the fitted profile so direct-saturation wings stop hiding a smaller dip beside them. For locating a feature, not for quantifying it.
- **Fit two dips together** — fits the deepest dip and the best other candidate simultaneously, so the major dip's wings do not bias the minor depth. Reports both positions, their separation, and the minor dip's

share of total depth. That share is a **depth ratio, not a population**: dip depth also depends on exchange rate, saturation field, `D18` and relaxation. Comparable between experiments run under identical conditions; not a substitute for Bloch-McConnell.

- **Zooming** — scroll to zoom X about the cursor, shift-scroll for Y, double-click to reset. The range boxes are a readout and follow the wheel; **Full range** also resets. The direct-saturation dip runs to near zero, so the baseline where a bound-state dip would sit is compressed — zooming Y to roughly 0.95–1.02 is usually what makes one visible.
- **Plot offset in ppm** — switches the x axis between ppm and Hz. The range boxes and any vertical markers are converted with it.

## Choosing the peak that is measured

The Z-spectrum is built from ONE window of the pseudo-2D rows: every row is measured over **Peak centre ± Half-width**, and that value divided by $I_0$ is one point.

- **Type it** — enter the shift in **Peak centre** (e.g. `-132.975`), set **Half-width (±)**, check the **Window** readout (range and point count), then click **Build Z-spectrum**. Typing does not rebuild by itself, and **Find peak** would overwrite your value with the peak nearest the carrier.
- **Pick from the list** — the **Resonances** list shows the peaks found in the least-saturated row. Double-click one to make it the main peak.
- **Picking with the mouse** on the 1D rows is not available yet; the panel does not display the rows themselves.

## Comparing several CEST experiments on one plot

Each overlaid curve is normalised to its OWN $I_0$, so experiments with different gain, scans or concentration remain comparable. Entries are labelled with the experiment, saturation field (CNST25) and D18.

1. **Open experiment…** and choose the first CEST expno folder (the numbered folder, not its parent), then **Build Z-spectrum**.
2. Right-click the plot → **Add CEST experiment to plot…**, or use the **Add experiment…** button under *Compare (overlaid Z-spectra)*. Choose the second expno folder.
3. Repeat for more. **Remove** drops the selected entries, **Clear all** removes them all; **Keep on plot** keeps the current curve before you load another experiment in its place.

Shortcuts: select several expnos in the browser and open CEST (the first is loaded, the rest overlaid), or drag expnos from the browser onto the CEST window.

Added experiments are measured with the CURRENT half-width, measurement, reference and error-bar settings, on the peak nearest the current main peak. If one cannot be added, the reason ("not a CEST experiment", "no FQ1LIST", "could not load" — usually the wrong folder) appears in the notes box at the top of the window.

Use this for the power series (30/60/100 Hz) and the partner-free control from Part 7 of the 19F CEST manual — a real exchange feature moves predictably with field and is absent from the control, and both are far easier to judge on one axis. The x axis is the offset from EACH experiment's own carrier, so compare experiments recorded with the same O1P.

## Right-click menu, markers and saving

Right-click either CEST plot (on a Mac: two-finger click or Control-click). The menu matches the main canvas:

- **Save image…** — same formats and writer as the main canvas (PNG, JPEG, TIFF, SVG, PDF, EPS, PS, PowerPoint; 300 dpi raster), exactly as zoomed. Also the **Save image…** button in the header, Ctrl+S / Cmd+S.
- **Copy image** — the same picture on the clipboard, to paste elsewhere.
- **Full range** (Z-spectrum) — back to the whole data range.
- **Add vertical / horizontal marker here**, **Remove nearest marker**, **Clear all markers** — pinned in data units and labelled with their value (ppm or Hz, $I/I_0$, ms), kept through zoom, rebuilds, overlays and saving, styled from Preferences like the main canvas.
- **Add CEST experiment to plot…**, **Keep current Z-spectrum on plot**, **Remove all overlays** (Z-spectrum).

A **crosshair** follows the pointer on both CEST plots, with its x and y values printed at the frame, as on the main canvas. It is not saved in images.

Scroll the wheel over the plot to zoom x about the cursor; hold Shift to zoom y; **drag with the left button to move the view**; double-click to reset. The range boxes follow.

## Carrier (O1P) and the offset axis

The x axis is the saturation offset from the **carrier**, i.e. the transmitter position O1P, not from the observed peak. The carrier in ppm is computed as (SFO1 - SF)/SF, so it stays correct after re-referencing with SR. Offsets are converted between Hz and ppm with SFO1. A check that both are right: the direct-saturation dip must be fitted at the peak's own offset from the carrier (shown in the notes box and the resonance list).

## Working with several resonances

**Find peak** picks the resonance nearest the carrier — the one the saturation offsets are measured from, and normally the one you want. The **peak list** below shows everything found, with each peak's offset from the carrier. The main peak is marked and always measured; tick another to add its Z-spectrum to the plot as a dashed curve, or double-click to make it the main peak instead. **Peak centre** stays freely editable if the detector misses one (see *Choosing the peak that is measured*).

Note that the carrier and the observed peak are different things. The peak is where you measure intensity; the carrier is the zero of the saturation axis. If they differ, the direct-saturation dip appears off-centre — a 0.137 ppm difference put it at +78 Hz on the test data. Setting O1P exactly on the resonance avoids this.

**detect above N x noise** controls how many peaks are listed. There is no universal setting: on one sample the main peak was 55x noise and a plausible second resonance only 6x. Lower it to look for weak resonances, raise it when the list fills with baseline ripple.

## Hunting a bound-state dip

1. Shift-scroll to about 0.95–1.02 in y so the baseline fills the plot.
2. Tick **Subtract fitted dip** to flatten the direct-saturation wings.
3. Read the candidate list. Three sigma is worth following up, five is convincing.

4 Confirm it: a real exchange feature persists in a repeat, moves predictably with saturation field, and is absent from a partner-free control. Direct saturation sits on the carrier; a bound state does not.

If nothing clears the threshold, the limit is noise rather than the tool. More scans, longer `D18`, or higher concentration lower the sigma figure, and the candidate list will follow.

This locates dips; it does not measure exchange. Turning a centre and depth into kex and a populated fraction needs Bloch-McConnell fitting against several saturation fields, which is not attempted.